

42 – Model Validation

Към момента имаме възможност за създаването на нови служители без попълнени данни. За да се избегне това, се изисква валидация.

ASP.NET Core Model Validation

The screenshot shows a web application interface with a 'List' and 'Create' button at the top. Below, there are two red error messages: 'The Name field is required.' and 'The Office Email field is required.' The form contains three input fields: 'Name' (with a red error message below it), 'Office Email' (with a red error message below it), and 'Department' (a dropdown menu with 'Please Select' and a plus icon). A blue 'Create' button is at the bottom.

За да направим поле Name, което е свързано с Employee.Name задължително, е необходим **[Required]** attribute, който съответно да извика namespace

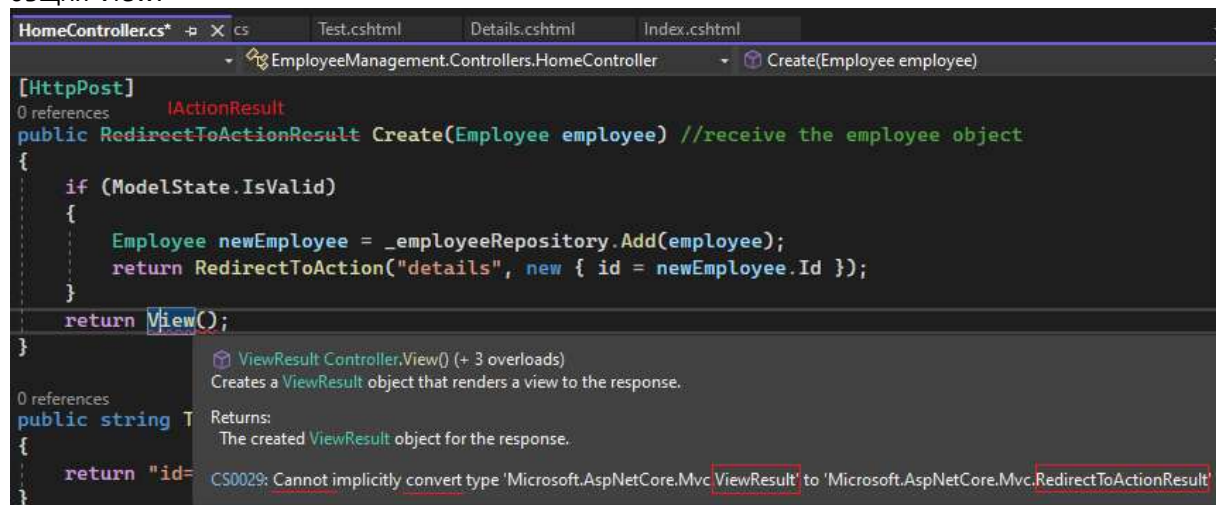
The screenshot shows the Visual Studio IDE with the 'Employee.cs' file open. The code defines a namespace 'EmployeeManagement.Models' and a class 'Employee' with a property 'Id'. A red squiggly line under '[Required]' indicates an error. The error message in the bottom right says: 'CS0246 The type or namespace name 'RequiredAttribute' could not be found (are you missing a using directive or an assembly reference?)'. The solution is to add the using directive 'using System.ComponentModel.DataAnnotations;'.

Атрибутите винаги предхождат обекта, за който се отнасят>

The screenshot shows the Visual Studio IDE with the 'Employee.cs' file open. The code defines a namespace 'EmployeeManagement.Models' and a class 'Employee' with properties 'Id', 'Name', 'Email', and 'Department'. The '[Required]' attribute is added to the 'Name' and 'Email' properties. Red boxes highlight the '[Required]' attribute and the 'Name' and 'Email' properties, with red arrows pointing from the attribute to the properties.

На този етап, ако се попълни празно поле Name, неговата стойност ще бъде Null и няма да се изпълни валидацията.

За да провери контролера дали валидацията е успешна или не, всичко, което трябва да направим, е да добавим if – ако валидацията е успешна, добавя нов служител, ако не, връща същия View.

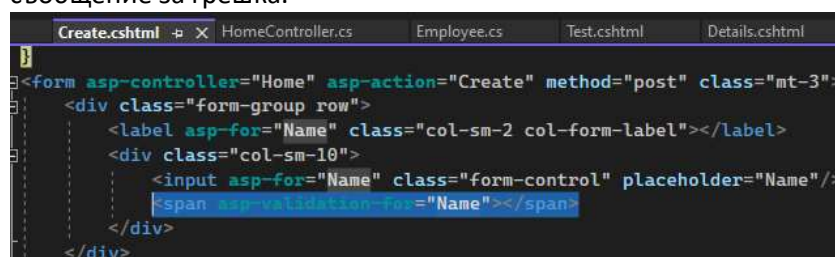


```
[HttpPost]
0 references
public IActionResult Create(Employee employee) //receive the employee object
{
    if (ModelState.IsValid)
    {
        Employee newEmployee = _employeeRepository.Add(employee);
        return RedirectToAction("details", new { id = newEmployee.Id });
    }
    return View();
}
0 references
public string T
{
    return "id="
}
```

```
[HttpPost]
public IActionResult Create(Employee employee)
{
    if (ModelState.IsValid)
    {
        Employee newEmployee = _employeeRepository.Add(employee);
        return RedirectToAction("details", new { id = newEmployee.Id });
    }
    return View();
}
```

* IActionResult = RedirectToActionResult + ViewResult

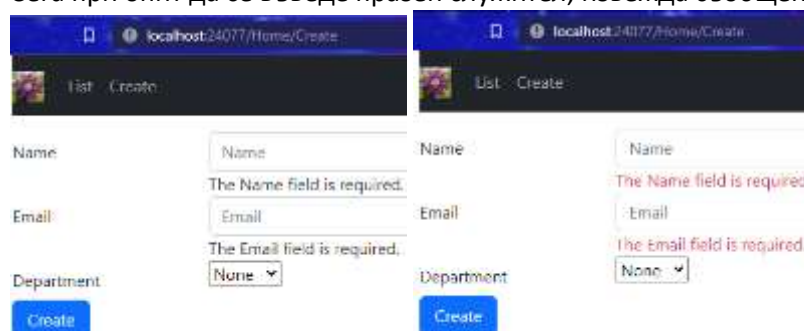
Остава да се изведе съобщение за грешка чрез C# asp-validation-for. Ако валидацията не се изпълни за Name property, на мястото на позиционирането на таг ще се изведе съобщение за грешка.



```
<form asp-controller="Home" asp-action="Create" method="post" class="mt-3">
<div class="form-group row">
<label asp-for="Name" class="col-sm-2 col-form-label"></label>
<div class="col-sm-10">
<input asp-for="Name" class="form-control" placeholder="Name"/>
<span asp-validation-for="Name"></span>
</div>
</div>
```

```
<span asp-validation-for="Name"></span>
<span asp-validation-for="Email"></span>
```

Сега при опит да се въведе празен служител, извежда съобщения за грешки.



Field	Value	Error Message
Name		The Name field is required.
Email		The Email field is required.
Department	None	

`class="text-danger"` прави текста червен.

`<span asp-validation-for="Name" class="text-danger"></span>`

A screenshot of a web form titled 'List Create'. It contains three input fields: 'Name' (empty), 'Office' (containing 'ABC123'), and 'Email' (containing 'Invalid email format'). Below the 'Email' field is a dropdown menu for 'Department' with the text 'Please Select'. A red error message 'The Name field is required.' is displayed below the 'Name' field, and another red error message 'Invalid email format' is displayed below the 'Email' field. A blue 'Create' button is at the bottom left.

Built-in Validation Attributes

- `RegularExpression`
- `Required`
- `Range`
- `MinLength`
- `MaxLength`
- `Compare`

Към момента обаче, форматът на Email не е валиден.

`[RegularExpression(@"^[a-zA-Z0-9_.+-]+@[a-zA-Z0-9-]+\.[a-zA-Z0-9-.]+$")]`

```
[Required]
[RegularExpression(@"^[a-zA-Z0-9_.+-]+@[a-zA-Z0-9-]+\.[a-zA-Z0-9-.]+$")]
7 references
public string Email { get; set; }
```

Резултат при грешно попълване на email:

A screenshot of a web form with three input fields: 'Name' (empty), 'Email' (containing 'AbC1'), and 'Department' (a dropdown menu with 'None' selected). A red error message 'The Name field is required.' is displayed below the 'Name' field, and another red error message 'The field Email must match the regular expression '^[a-zA-Z0-9_.+-]+@[a-zA-Z0-9-]+\.[a-zA-Z0-9-.]+\$'.' is displayed below the 'Email' field. A blue 'Create' button is at the bottom left.

За да променим съобщението, добавяме Property `ErrorMessage = "Invalid email format".`

```
[RegularExpression(@"^[a-zA-Z0-9_.+-]+@[a-zA-Z0-9-]+\.[a-zA-Z0-9-.]+$",
ErrorMessage = "Invalid email format")]
```

Към момента, текстът Email е автоматично прихванат, но ако желаем да го променим на Office email, използваме друг атрибут `[Display (Name="Office Email")]`

A screenshot of a web form with three input fields: 'Name' (empty), 'Office Email' (containing 'AbC1'), and 'Department' (a dropdown menu with 'None' selected). A red error message 'The Name field is required.' is displayed below the 'Name' field, and another red error message 'Invalid email format' is displayed below the 'Office Email' field. A blue 'Create' button is at the bottom left.

Последно искаме да допуснем максимална дължина от 50 символа за името с персонализирано съобщение за грешка.

`[MaxLength(50, ErrorMessage = "Name cannot exceed 50 charakters")]`

При мен обаче така и не извежда подобна грешка, просто не допуска да надхвърля при самото въвеждане повече от 50 символа. Когато опитам ограничение с 60 символа, по същият начин не ми позволи да ги надхвърля още при въвеждането.

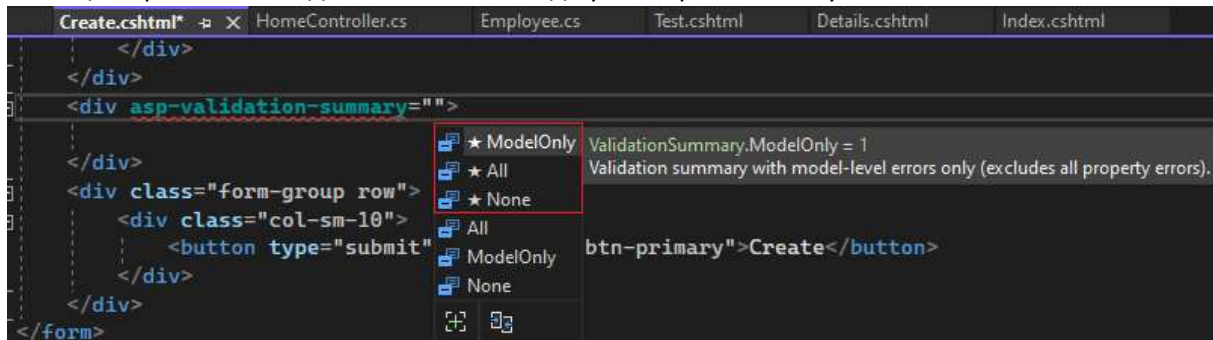
```
public class Employee
{
    public int Id { get; set; }
    [Required]
    [MaxLength(60, ErrorMessage = "Name cannot exceed 50 charakters")]
```

```

public string Name { get; set; }
[Required]
[RegularExpression(@"^[a-zA-Z0-9_+]+@[a-zA-Z0-9]+\.[a-zA-Z0-9-]+\.$",
    ErrorMessage = "Invalid email format")]
[Display(Name="Office Email")]
public string Email { get; set; }
public Dept Department { get; set; }
}

```

Към момента под всяко поле за въвеждане се извежда съответната грешка, но може да обобщим грешките на едно място точно над бутона чрез summary:



```
<div asp-validation-summary="All"></div>
```

Name
The Name field is required.

Office Email
Invalid email format

Department

- The Name field is required.
- Invalid email format

съответно отново може да променим цвета до червен с `class="text-danger"`.

ASP.NET Core Model Validation

Step 1 : Apply Validation Attributes on Properties

```

public class Employee
{
    public int Id { get; set; }
    [Required, MaxLength(50)]
    public string Name { get; set; }
    [Display(Name = "Office Email")]
    [RegularExpression(@"^[a-zA-Z0-9_+]+@[a-zA-Z0-9]+\.[a-zA-Z0-9-]+\.$",
        ErrorMessage = "Invalid email format")]
    [Required]
    public string Email { get; set; }
    public Dept Department { get; set; }
}

```

Step 2 : Use **ModelState.IsValid** property to check if validation has failed or succeeded

```
[HttpPost]
public IActionResult Create(Employee employee)
{
    if (ModelState.IsValid)
    {
        Employee newEmployee = _employeeRepository.Add(employee);
        return RedirectToAction("details", new { id = newEmployee.Id });
    }

    return View();
}
```

Step 3 : Use **asp-validation-for** and **asp-validation-summary** tag helpers to display validation errors

```
<div asp-validation-summary="All">
</div>

<div>
    <label asp-for="Name"></label>
    <div>
        <input asp-for="Name">
        <span asp-validation-for="Name"></span>
    </div>
</div>

<div>
    <label asp-for="Email"></label>
    <div>
        <input asp-for="Email">
        <span asp-validation-for="Email"></span>
    </div>
</div>
```

Validation: I Attributes [Required]

If check: II ModelState.IsValid

Errors: III asp-validation-for / -summary